

Total No. of Questions : 8]

SEAT No. :

PE-2596

[Total No. of Pages : 4

[6583]-127

T.E. (Information Technology)
THEORY OF COMPUTATION
(2019 Pattern) (Semester - V) (314441)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Attempt Q.1 or Q.2, Q.3. or Q.4, Q.5 or Q.6, and Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Write a formal definition of a Regular Grammar? Explain Chomsky Hierarchy of Grammar. [6]

b) Convert the following Left Linear Regular Grammar (LRG) into Right Linear Regular Grammar (RLG). [6]

$G = \{V, T, S, P\}$ where $V = \{S, A, B, C\}$, $T = \{0, 1\}$, $S = \{S\}$ and P are given as

$S \rightarrow B1 \mid A0 \mid C0$

$B \rightarrow B1 \mid 1$

$A \rightarrow A1 \mid B1 \mid C0 \mid 0$

$C \rightarrow A0$

c) Derive the leftmost and rightmost derivation for the string "ibtibtaea" using following Context Free Grammar (CFG). [6]

$S \rightarrow iCtS \mid iCtSeS \mid a$

$C \rightarrow b$

OR

P.T.O.

Q2) a) Simplify the following CFG (Context Free Grammar) [6]

$S \rightarrow aSa \mid bSb \mid \varepsilon \mid A$

$A \rightarrow aA \mid \varepsilon$

$D \rightarrow b \mid bD \mid \varepsilon$

b) Convert the CFG (Context Free Grammar) with productions [6]

$S \rightarrow bA \mid aB$

$A \rightarrow bAA \mid aS \mid a$

$B \rightarrow aBB \mid bS \mid b$

to Chomsky Normal Form (CNF).

c) Write a short note on the Pumping Lemma for Context Free Language. [6]

Q3) a) Construct a Push Down Automata (PDA) for accepting language containing odd length palindrome over $\Sigma = \{0, 1\}$. [6]

b) Give the Context Free Grammar (CFG) generating the language accepted by the following Pushdown Automata (PDA) [6]

$M = (\{p, q\}, \{0, 1\}, \{Z_0, X\}, \delta, q, Z_0, \phi)$ when δ is given by

$\delta(q, 1, Z_0) \rightarrow (q, XZ_0)$

$\delta(q, 1, X) \rightarrow (q, XX)$

$\delta(q, 0, X) \rightarrow (p, X)$

$\delta(q, \epsilon, X) \rightarrow (q, \epsilon)$

$\delta(p, 1, X) \rightarrow (p, \epsilon)$

$\delta(p, 0, Z_0) \rightarrow (q, Z_0)$

c) Define the following terms with example. [6]

i) Pushdown Automata

ii) Post Machine

OR

Q4) a) Construct a Push Down Automata (PDA) that accepts the following language. $L = \{ a^n b^n \mid n \geq 0 \}$ [6]

b) Design a Post Machine which accepts the string with an equal number of a's and b's. [6]

c) Construct a Push Down Automata (PDA) for a language $L = \{ a^n b^{2n} \mid n \geq 0 \}$ [6]

Q5) a) Construct a Turing Machine (TM) for checking if a set of parenthesis are well formed. [6]

b) Write a short note on : [6]

i) Church-Turing thesis

ii) Multi-tape TM

c) Explain the power of Turing Machine over Push Down Automata. [5]

OR

Q6) a) Design a TM to compute proper subtraction of two unary numbers. The proper subtraction function f is defined as follows. [6]

$$\begin{aligned} f(m,n) &= m - n && \text{if } m \geq n \\ &= 0 && \text{otherwise} \end{aligned}$$

b) Construct a Turing Machine to accept a language $L = \{ a^{n+1} b^n \mid n \geq 0 \}$ [6]

c) Write a short note on Universal Turing Machine. [5]

Q7) a) Write Short note on : [6]

i) Un-decidable Problem

ii) Measuring Complexity

b) Let $HALT_{TM} = \{ \langle M, w \rangle \mid M \text{ is a TM and } M \text{ halts on input } w \}$. Prove that $HALT_{TM}$ is undecidable. [6]

c) Define the following. [5]

i) NP Hard Problem

ii) NP Complete Problem

OR

- Q8)** a) Prove that A_{DFA} is a decidable language. [6]
 $A_{DFA} = \{ \langle B \rangle \mid B \text{ is a DFA that accepts input string } w \}$.
- b) Write a short note on Normal Forms of Boolean Expressions. [6]
- c) Write a short note on NP Completeness of SAT Problem. [5]
